

# Stroke Amnesia: Why Language Models Still Need a Canvas

Poonam Nair<sup>1</sup>  
May 2026

## Abstract

Large language models can describe drawings fluently and emit SVG, Canvas, or TikZ code that renders to recognisable images. We argue these abilities do not constitute robust drawing competence. Inky is a coding-agent-driven canvas animation pipeline that turns user-provided storyboard references into hand-drawn, code-rendered 2D animations. The system is a scaffolding and rendering harness rather than a fixed set of examples: each project defines its own canvas dimensions, frame count, and a scaffold of intermediate artifacts including requirements, construction blueprints, in-between plans, and renderer stubs. Through this generic pipeline, we observe consistent failure patterns — *stroke amnesia*, *compositional collapse*, and *patch reflex* — and show how externalised structure, visual feedback, and root-cause redraw rules push outputs back toward coherence. We argue that the missing capability is not pixel synthesis but a durable latent canvas that tracks object identity, part ownership, and contact across generation and revision. We close with an evaluation agenda that grades *process*, *structure*, and *repair behaviour*, independent of final image content, and three benchmark tasks designed to discriminate models on these axes.

**Keywords:** large language models, drawing, multimodal evaluation, agentic systems, generative art, benchmarks.

## Introduction

A model that can describe a hand is not yet a model that can keep the wrist attached. This paper is a position statement on a gap that has become visible only recently, as multimodal models, tool use, and agentic coding loops have made it plausible for language models to act through canvases, SVG, browsers, and visual feedback. They can now *try*, *inspect*, and *revise*. We argue that this revision loop is where the gap appears most sharply: models can write code that produces drawings, and they can describe what is wrong with the result, but they cannot reliably repair the result without external structure.

The argument is grounded in INKY, a coding-agent-driven canvas animation pipeline that turns user-provided storyboard references into hand-drawn, code-rendered, inspectable 2D animations. INKY is not a research artefact and not a fixed showcase; it is a scaffolding and rendering harness. Given a storyboard, it creates a project folder, generates intermediate artifacts (requirements, construction blueprints, in-between plans, renderer stubs), and hands a coding agent a structured workspace to fill in. The project’s canvas dimensions, frame count, and material kit are defined per project rather than fixed by the system. Its design rules, accumulated across many user projects, document the failure modes that matter in practice: floating limbs, disconnected clothing, mistaken speech-bubble ownership, guide marks promoted to final objects, and texture used to hide unclear construction. We treat these rules as evidence.

We make three contributions.

1. A failure taxonomy for LLM-mediated drawing, grounded in a production pipeline rather than synthetic prompts, including three named phenomena: *stroke amnesia*, *compositional collapse*, and *patch reflex*.
2. An engineering pattern for storyboard-to-canvas animation using explicit intermediate artifacts (requirements, lighthouses, construction blueprints, semantic review boards, visual diffs) that survives model swaps.
3. A benchmark agenda that grades drawing as a process — with decomposed scoring across recognisability, spatial relation, part attachment, text readability, repair quality, and continuity — and three concrete tasks designed to be hard for today’s models in informative ways.

The paper is deliberately constructive. We critique the implicit claim that LLMs “can draw” because they can emit Canvas or SVG snippets, but the centre of gravity is the pipeline and the benchmark, not the critique.

## Background and Scope

### What we mean by “drawing”

In this paper, drawing is a *process trace that places visual marks on a 2D canvas so that objects, bodies, text, motion, and material texture are readable without the source prompt*. In INKY, this includes storyboard redraws, comic frames, captions, speech bubbles, hand-drawn texture, and frame-by-frame animation. We explicitly exclude photorealistic text-to-image diffusion, 3D modelling, music notation, handwriting as a separate recognition problem, and claims about final production art quality. Our concern is structural readability of 2D illustrative drawing produced through code.

### What we mean by “bad”

We measure “bad” relative to two reference points: (i) the model’s own language competence on the same subject, and (ii) a competent human sketcher working under similar time pressure. The failure is not primarily aesthetic. It is structural: parts detach, proportions drift, intended objects become blobs, and repairs are made with patches rather than rebuilt geometry. A model that can write a correct paragraph describing a person holding a mug can produce a rendering in which the hand floats next to the cup.

### Why now

Three shifts make this paper timely. First, multimodal models can now see their own output, which makes the gap between description and execution observable in a single conversation. Second, agentic coding loops allow models to act, render, inspect, and revise — the ingredients of drawing as practice rather than synthesis. Third, the external

rendering stack (browser canvases, headless screenshots, visual diff libraries) is now cheap enough to wire into a tight loop. The bottleneck has moved from *can the model emit drawing code?* to *can the model repair drawing structure when shown that it is wrong?*

## The INKY Pipeline

### System overview

INKY renders 2D animations from user-provided storyboard references using a coding agent (e.g. Claude or a similar code-writing model) that operates over a structured repository of design rules, skills, and project files. The runtime application itself does not call a model API; the “drawing brain” is the agent operating on the codebase. The user-facing artefact is a browser preview with Play/Pause, timeline scrubbing, speed control, PNG and MP4 export.

Canvas dimensions, frame count, frame rate, and material kit are defined per project, not by the system. A typical project might use a  $960 \times 620$  canvas at 12 FPS over roughly 96 frames, but each parameter is configurable through the project’s `requirements.md` and `renderer` stub. The drawing primitive is JavaScript Canvas: paths, ellipses, fills, text, material strokes, procedural texture, and frame schedules. Helper functions wrap these into material strokes, speech bubbles, captions, body parts, props, and animated scenes.

### The end-to-end agent pipeline

Given a storyboard image and a one-line user prompt (e.g. “animate a cat playing with yarn in ink and watercolour style”), INKY expects the agent to work through a ten-step pipeline: (1) create a new folder under `projects/<project-name>/` (typically via `npm run storyboard:start`); (2) save the original storyboard in `projects/<project-name>/image/`; (3) extract the individual panels into `projects/<project-name>/storyboard/` (`npm run extract:storyboard`); (4) write `requirements.md` with the requested style and structural notes; (5) generate construction blueprints from the source panels (`npm run storyboard:blueprint`); (6) redraw the scene in canvas code using the storyboard as a lighthouse,

not a tracing source; (7) remove storyboard-only marks such as panel numbers, borders, long guide arrows, and construction lines; (8) animate between the panels with in-between frames (`npm run storyboard:inbetween`); (9) run polish checks for body connections, clothing clarity, props, speech bubbles, texture, and flicker (`npm run storyboard:polish`, `npm run storyboard:visual-diff`, `npm run storyboard:semantic`); and (10) verify the browser preview and export controls.

This end-to-end pipeline is distinct from the nine-step *drawing order* introduced later in §5.3; the agent pipeline covers project creation, scaffolding, and review, while the drawing order governs how a single frame is constructed inside that scaffold.

### Why Canvas, not SVG or diffusion

Canvas is deterministic, browser-native, easy to capture frame by frame, and compatible with procedural brush systems. INKY avoids pasting source images or relying on diffusion output because the point is *controllable construction and editable animation*. The representation chosen is not a stylistic preference; it is a commitment to a substrate where every mark has a code-level cause.

### The effective prompt

The agent’s effective prompt is not a single string. It is a set of files: an `AGENTS.md` that defines obligations, material tools, failure modes, and quality gates; a `DESIGN.md` that defines aesthetic and structural rules; per-skill `SKILL.md` files for specialised procedures (`storyboard-animation-pipeline`, `reference-lighthouse`, `frame-construction-planner`, `inbetween-frame-planner`, `semantic-clarity-auditor`, `frame-consistency-auditor`, `animation-polish-pass`, `caption-subtitle-tooling`, and `comic-speech-bubble-tooling`); and a per-project `requirements.md` that captures the brief. Context is externalised into the filesystem, so the agent re-reads the relevant artefact rather than carrying everything in prompt memory.

### Project structure

Each render project occupies a fixed folder layout:

```
projects/<project-name>/
  image/      # original source image
  prompt/     # user directions, corrections
  storyboard/ # frames, requirements,
              # ledgers, blueprints,
              # in-between plans
  outputs/    # PNG frames, MP4, contact
              # sheets, exports
```

The structure is load-bearing rather than cosmetic: the agent is instructed not to copy old experiments, scratch files, or unrelated outputs into a new project folder, because cross-project contamination is itself a documented failure mode.

The skill files under `skills/` follow the open Agent Skills format (a folder per skill containing a `SKILL.md` with YAML frontmatter and prose instructions, with optional

bundled resources). The format was originally developed by Anthropic and adopted as an open standard; INKY’s skills are portable in principle, not bespoke to one model provider.

### Repository layout

For readers consulting the repository alongside this paper, the mapping is straightforward. `AGENTS.md` and `DESIGN.md` at the repository root define the load-bearing prompt material discussed in §3.4. The `skills/` folder contains the per-skill `SKILL.md` files referenced throughout. The `src/`

folder contains the runtime: `material-tools.js` (brush kit and pressure profiles), `rough-accent-tools.js` (guarded RoughJS), `speech-bubble-tools.js` (bubble construction and ownership validator), and `caption-tools.js`. Pipeline scripts live under `tools/` as a series of Node `.mjs` files, each surfaced as an `npm run storyboard:*` command (scaffolding, blueprint generation, in-between planning, polish pass, visual diff, semantic review, material preview). Per-project artifacts accumulate under `projects/`.

### **Material tools**

Rather than a single “drawing” API, INKY exposes a kit of roughly two dozen named brush helpers in `src/material-tools.js`. Each entry pairs a stroke configuration (size, thinning, smoothing, jitter, alpha, hatch length) with a pressure profile and a written `useWhen` clause that tells the agent when the brush applies: `technical-pen` for crisp outlines and prop edges, `dip-ink` or `brush-pen` for expressive silhouettes and hair, `ballpoint-pen` for scratchy doodles, `marker` for translucent fills, `graphite-pencil` and `colored-pencil` for construction and dry colour, `wax-crayon` and `oil-crayon` for wax-gap texture, `charcoal` for smoky shadow, `watercolor`, `ink-wash`, `salt-watercolor`, `gouache`, `acrylic`, and `oil-paint` for paint effects, and `airbrush`, `sponge`, and `screen-tone` for specific effects. Two auxiliary helpers do work that prompts alone cannot reliably do: `ensureReadableColor()` adjusts text-bearing shapes against varying backgrounds with a contrast target and returns whether the colour had to be adjusted, and `drawCoherentPaperGrain()` produces seeded paper tooth that is stable across frames instead of random per-frame speckle. The paper-grain function is itself a continuity countermeasure: shimmer in locked backgrounds is one of the visual-diff stage’s named targets.

The kit acts as a soft DSL. It constrains the action space at the exact point the model is most likely to drift — choosing “hand-drawn” as a style word and emitting noisy texture rather than committing to a medium with structural implications. The kit is also defensive: the RoughJS accent helper exposes explicit arrays `ROUGH_SAFE_USES` and `ROUGH_AVOID_USES` in code, restricting it to non-semantic accents (background texture, dust, sparkle, paper edges) and excluding speech bubbles, readable text, labels, borders, desks, frames, hands, faces, body silhouettes, and story-critical contact points. The constraint is encoded as data, not only as text in a prompt.

### **Non-LLM components**

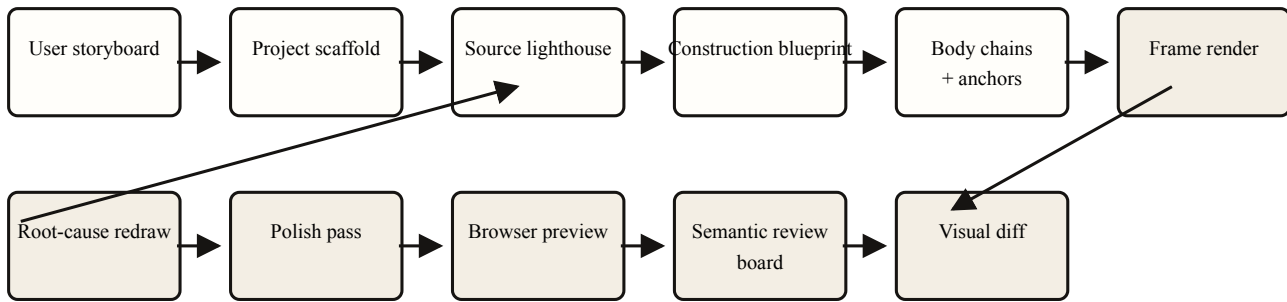
Several components in the loop are deterministic tools rather than model calls: storyboard extraction, project scaffolding, construction blueprint generation, in-between schedule generation, speech-bubble and caption-track generation, frame rendering, visual diffing, semantic review board generation, browser preview, and MP4 assembly. The runtime stack is Vite over vanilla HTML/CSS/JS with Canvas 2D for drawing. Node scripts (one per pipeline stage) live under `tools/`, each invoked through an `npm run storyboard:*` command. Production dependencies are deliberately small: `perfect-freehand` for pressure-aware stroke generation, `chroma-js` for colour operations, `simplex-noise` for seeded texture noise, `d3-ease` for animation curves, `pixelmatch` and `pngjs` for pixel-diff review, `roughjs` for guarded non-semantic accents, `paper` as an optional vector-graphics layer, `jszip` for project bundling, and `mp4-muxer` as a deprecated MP4 path scheduled for replacement with `Mediabunny`. `ImageMagick` is used as a system tool for contact-sheet composition and is not an npm dependency.

### **Quality gates**

Before final export, the pipeline requires the following checks: source-as-lighthouse with anchors, contact points, and silhouettes identified; every important body chain connected; clothing and limbs forming plausible parent-child chains; recurring props visible when the scene implies them; chosen brush kit supporting the look without hiding unclear construction; storyboard-only guide marks absent from the final canvas; a polish-pass run after the first render; root-cause redraw rather than cover-up patches; in-between frames inspected, not only key frames; visual-diff review for flicker, locked-background drift, vanishing props, and bubble jumps; and browser controls (Play, Pause, speed, scrub, PNG and MP4 export) verified end-to-end. When captions or speech bubbles are present, CC toggle, VTT/SRT export, text wrapping, tail ownership, and bubble placement are checked explicitly.

### **Pipeline stages**

Figure 1 summarises the stages. A drawing is not “done” when the agent emits final code; it is done when an explicit review pass — browser preview, visual diff, export check, project-specific gates — accepts the frame sequence. There is no autonomous stop signal.



The INKY pipeline, drawn generically: any user storyboard becomes a scaffolded project that flows through construction (top row, left to right) and review (bottom row, right to left). Failures discovered in review trigger *root-cause redraw* from the construction blueprint, not patches over the rendered frame.

### Three Named Failure Modes

Across many projects, three failure modes recur often enough to deserve names. They are not mutually exclusive — a single defective frame often exhibits all three — but naming them makes prompts, review checklists, and benchmarks more precise.

#### *Stroke amnesia*

A model emits a stroke, then loses track of what the stroke meant. Earlier marks become ambiguous to its own attention: a guide arrow becomes a wire or a limb in the final rendering; a construction line becomes a decorative element; a speech-bubble tail tracks the wrong speaker. The repo guards against this with explicit instruction: panel numbers, borders, labels, long arrows, long jump or throw arcs, and construction lines must be removed or translated into actual animation timing, never left on the final canvas. The fact that this guard exists in writing is itself evidence: the agent reliably promotes these marks to final objects when the instruction is absent.

The root cause is not a missing concept of “arrow versus wire”; the model can explain the difference. The cause is the lack of durable ownership: there is no inspectable record stating that *this mark belongs to construction, not to the final scene*.

We call this *stroke amnesia*. The signature is that the model will defend the offending mark when asked, because by the time it is asked, the mark has been reabsorbed into “what the drawing is.”

#### *Compositional collapse*

The model can draw a cat. It can draw a kettle. Asked to draw a cat to the left of a kettle on a table, with the cat watching steam rise, relations begin to fail: the cat sits on top of the kettle, the kettle floats above the table, or the steam emerges from the cat. Composition breaks when several relations must hold at once: a character talks while holding papers, a bubble must avoid the face, a prop must remain on the desk, and the same object must stay visible across frames.

We call this *compositional collapse*: the rate at which relational constraints are satisfied appears to fall non-linearly as the number of simultaneous constraints rises. The phenomenon is familiar from generative image models; what is new is that an agentic loop with vision feedback does not, on its own, fix it.

#### *Patch reflex*

Shown a defect, the model often repairs the rendered output rather than the underlying construction. The repo enumerates the specific cover-ups it has had to forbid: cover-up shapes, eraser strokes, white patches, opacity tricks, and decorative texture. Patches succeed locally and fail under animation, when the camera moves, or when the defect recurs across frames. A particular case worth naming is the speech-bubble tail: the repo treats visible circles, ovals, socket rings, and dot trails as bugs — those marks belong only to thought bubbles, and the speech-bubble body and tail must be constructed as one continuous closed path, then filled and stroked once.

We call this *patch reflex*. It is the drawing analogue of a common failure in code generation: editing the symptom rather than the cause. INKY’s most load-bearing rule is therefore not stylistic but procedural — the named *Root-Cause Redraw Rule*: when a rendered frame has a bug, fix the cause and rebuild the affected shape or layer from its construction anchors using the source lighthouse, requirements, and neighbouring frames, then rerender. Cover-ups are not permitted.

#### *Why these three matter*

Stroke amnesia attacks identity. Compositional collapse attacks relation. Patch reflex attacks repair. Each maps to a missing capability rather than a missing skill. A larger model with no other changes will likely improve planning and code syntax, but we expect gains on these three axes to plateau without (i) durable representations of object identity and (ii) training pressure on repair behaviour, not only on final images.

### Workarounds in INKY

INKY’s pipeline is, in effect, a catalogue of engineering workarounds for the three failures above. We summarise the ones that have proven most durable, with a note on which

we believe survive model swaps.

### Source lighthouse and construction anchors

Before detailed rendering, the agent produces a *source lighthouse*: an interpretation of the storyboard reference that names body chains, contact points, risky shapes, and storyboard-only marks. These become construction anchors that subsequent stages must respect. The lighthouse is explicitly not a copy of the source; it is a structural reading of it. This single workaround produced the largest practical quality jump we observed.

The lighthouse is also externalised as imagery, not only text. A dedicated tool, `make-construction-blueprints.mjs`, takes each source frame and uses ImageMagick to compose a board that joins the full source on the left with a vertical strip of semantic crops on the right (face, hands, action props). The board is paired with a markdown file naming attachment chains, risky shapes, the silhouette gate, and the detail-lock rule. The agent therefore consults the lighthouse not as a paragraph of text but as a rendered, croppable, inspectable artefact that survives across sessions.

### Parent–child attachment chains

A specific construction discipline does much of the work against *compositional collapse*: every body and costume follows a parent–child chain. Clothing is not floating decoration. Torso owns the shirt hem; the shirt hem overlaps the waistband; the waistband owns the leg openings; the leg openings own the legs. When the pose shifts, bends, or rotates, the clothing anchor moves with the body rather than as an independent shape. The repo enforces this with explicit examples: if the source specifies “shorts,” the agent must draw a waistband, two leg panels or openings, a centre seam or crotch split, and legs exiting underneath. A single rounded blue shape is logged as a clarity failure, not a stylistic choice. The chain is the durable structure that survives re-rendering across frames.

The rule is not only written; it is mechanised. In a typical INKY renderer, the function `shortsGeometryForPose` computes the waistband, leg edges, and openings from the torso geometry, and `anchorLegChainsToShorts` re-anchors the leg chains to those openings before any limb is drawn. The renderer then uses `straightMaterialShape` — not the smoothed closed-path variant — so the leg panels render as four-sided shapes with a visible centre seam rather than a single rounded blob. The structural rule and the structural code are the same artefact:

```
const leftLeg = [
  [leftEdge, waistBottomY - 1],
  [centerX - 4, waistBottomY],
  [centerX - 5, legBottomY - 2],
  [leftEdge + 12, legBottomY + 3],
  [leftEdge - 5, waistBottomY + 20]];
const rightLeg = [
  [centerX + 4, waistBottomY],
  [rightEdge, waistBottomY - 1],
  [rightEdge + 5, waistBottomY + 20],
  [rightEdge - 12, legBottomY + 3],
  [centerX + 5, legBottomY - 2]];
const waistband = [
  [leftEdge - 4, waistTopY],
  [rightEdge + 4, waistTopY],
  [rightEdge + 2, waistBottomY],
  [leftEdge - 2, waistBottomY]];

straightMaterialShape(leftLeg, /*...*/);
straightMaterialShape(rightLeg, /*...*/);
straightMaterialShape(waistband, /*...*/);

// centre seam
line([centerX, waistBottomY],
  [centerX, legBottomY - 1], 1.9, ...);
```

Three points are worth surfacing. First, the openings are derived from the torso, so a pose change automatically moves them. Second, the legs are explicitly straight-edged closed paths rather than ellipses or rounded blobs, which prevents the documented “single rounded blue shape” failure mode at the geometry level. Third, the centre seam is a separate `line()` call, not a decorative afterthought — it is constructed in the same step as the panels, not patched in later.

### The nine-step drawing order

Decomposed layering is not a style preference; it is encoded as a strict construction order with nine steps: (1) paper and chosen material background, (2) simple construction

anchors, (3) body chains and prop contact points, (4) main silhouette, (5) removal or translation of storyboard-only guide marks, (6) ink contours, (7) dry colour, watercolour, gouache, pencil, crayon, charcoal, or other chosen material texture, (8) dense hatching and texture, and (9) a final clarity pass. The order matters because later stages cannot modify foundational structure without revisiting earlier ones, and because step 5 — explicit removal of guide marks — is the structural countermeasure to *stroke amnesia*.

The transition between step 4 (silhouette) and step 6 (ink) is guarded by an explicit *silhouette gate*, written into every construction blueprint the pipeline produces: body parts must be nameable before texture; there must be no blank gap

between attached body masses; hands must have wrists or cuffs; facial features must sit inside the head shape; props must read as separate objects before hatching and watercolour detail; and any user-requested changes must be visible as clean silhouettes. If the silhouette gate fails, the pipeline is instructed not to proceed to ink or texture — the canonical *detail lock*. This is what turns the construction order from a recommendation into an enforceable contract.

### ***The polish pass as a second pipeline***

INKY treats the first drawn frame or first full render as a draft. A separate procedure, *animation-polish-pass*, is invoked after each risky frame and always before final export. It is structured as nine ordered audits: (1) lighthouse replay, comparing each frame to its source for pose, silhouette, contact, scale, and text ownership; (2) gap closure for disconnected heads, wrists, sleeves, hands, hair, props, labels, shadows, and contact points; (3) artifact removal of accidental panel borders, frame numbers, long guide arcs, construction lines, and brush curves on rigid borders; (4) semantic audit, cropping risky zones and confirming they read without the source image; (5) consistency audit across adjacent frames for character identity, hair, face placement, clothing, props, and background stability; (6) motion audit for popping, floating features, vanishing props, or tails and captions drifting from the speaker; (7) visual-diff audit using pixel-diff sheets; (8) material enhancement, added only after form reads clearly; and (9) browser and export check.

The polish pass enforces a list of named checks that is, in effect, a deployed detector for the three failure modes: the speech-bubble tail points to the intended speaker; the bubble is constructed as one continuous body-plus-tail path with no background-coloured wedge or eraser seam; visible tail-socket circles, ovals, or thought dots are absent from speech bubbles; hair has a part, hairline, strand groups, or side locks rather than reading as a blob; rigid borders, labels, panels, desks, and nameplates are drawn with closed paths or straight segments rather than wandering brush arcs; clothing follows the body chain; recurring props do not vanish during key frames or in-betweens; and the final fix is a redraw of the broken source geometry or layer order, not a visible cover-up. These checks did not arrive as design wishes; they arrived as forensic responses to recurring failures.

### ***The in-between layer model***

A separable failure mode operates across frames rather than within them. Keyframe attachment is necessary but not sufficient: if the body chain is correct at frame  $k$  and at frame  $k + 8$ , the in-betweens at frames  $k + 1$  through  $k + 7$  can still detach. We treat this as *inter-frame attachment drift*, a sub-species of stroke amnesia that operates over the sequence.

INKY's in-between planner addresses this with two concrete mechanisms. First, layers are partitioned into three sets that the in-between schedule treats differently: *locked layers* (paper grain, panel border, wall notes, desk, static background lighting) which must not move across the

transition; *anchored layers* (laptop, mug, notebooks, papers, phone, lamp, plant, recurring desk props) which hold position unless the story moves them; and *animated layers* (head, torso, shoulders, arms, wrists, hands, facial expression, hair droop, motion marks) which interpolate between keyframes. Each in-between frame in the schedule carries a `keepAttached` array — [head-face-details, shoulder-sleeve-forearm-wrist-hand, props-contact-surface] — which forces the bridge frames to honour the same attachment chains as the keyframes.

Second, the planner emits *three timing channels* per in-between rather than one. Legacy `linearT` preserves even spacing; `motionT` is reserved for spatial anchor interpolation and uses the action-appropriate ease curve; and `easedT` is reserved for local accents (settle, bounce, anticipation). Easing is not chosen by the model. A deterministic `pairComplexity()` function reads the action text of each transition and returns a complexity score; the score selects from seven named profiles (bounce, anticipate-settle, settle, ease-in-out, ease-out, soft-bridge, linear). This removes a class of subtle motion bug — the everything-is-linear feel that LLM-emitted animation code tends to produce by default — at plan time, before any frame is rendered.

### ***Visual feedback as a first-class artefact***

Browser screenshots, rendered PNG frames, ImageMagick contact sheets, semantic crops, and pixel-diff reports are central to the loop. The visual-diff stage targets a specific class of defect that text-only inspection misses: shimmer in locked backgrounds, popping or vanishing props, drifting speech bubbles and captions, and frame-to-frame changes larger than the intended action. The expected gain is largest precisely for defects that are obvious to the eye but hard to infer from code text alone. The cost is render and review time; we consider it well spent.

A specific anti-shimmer technique deserves mention: the entire animation uses a small seeded pseudo-random generator (`mulberry32`) whose seed is derived from the frame number. Material jitter, paper grain, and procedural texture are all deterministic per frame. This is the structural answer to “shimmer in locked backgrounds” — not a smoothing filter, but a guarantee that two renders of the same frame produce the same marks.

### ***Runtime validators***

INKY also ships in-code validators that run at startup and log warnings before the first frame is rendered. The clearest example is `validateSpeechBubbleOwnership`: it walks the bubble track, computes the distance from each bubble's tail tip to the intended speaker's anchor at the relevant frame, and flags any case where the distance exceeds a project-defined threshold (typically on the order of 100–150 pixels for a ~960px canvas). Bubble-speaker drift, one of the canonical *stroke amnesia* signatures, is detected before any rendering takes place. The check is data, not a prompt: a new coding agent inheriting the project sees the warning array on first load.

### ***Root-cause redraw, not patch***

If only one workaround survived, it would be this. Patch-style repairs — white plugs, opacity tricks, texture cover, eraser seams — are explicitly rejected. The rule encodes the lesson that *patch reflex* is durable and dangerous: it produces results that look acceptable in a single frame and fail across the sequence.

### ***Externalised context as durable memory***

Inky externalises context into files: requirements, ledgers, blueprints, speech-bubble JSON, in-between plans, polish notes. The agent re-reads the relevant artefact instead of carrying everything in prompt memory. This is the cheapest substitute we have found for a durable latent canvas.

### ***A minimal external library surface***

A subtler workaround is restraint about which libraries the agent may reach for. INKY prefers the existing canvas frame renderer for final output and admits external libraries only against a clear missing capability: Anime.js for object-property timelines and seekable choreography, Motion for app UI transitions and gestures, Fabric.js for an interactive editor with selectable canvas objects, and Atrament for live pressure and smoothing in stroke capture. RoughJS is restricted to guarded non-semantic accents. The discipline matters because each additional library is also an additional surface on which the agent can hide structural problems under unfamiliar texture or animation.

### ***Which workarounds survive model swaps***

In our experience, persistent intermediate artefacts (requirements, blueprints, body chains, rendered frames, visual diffs) survive model swaps because they are external process structure. Handwritten prompt rules that depend on one model’s compliance are the most fragile: a new model may ignore a constraint like *do not use RoughJS for speech bubbles* unless the constraint is encoded in tools or in helper APIs that simply do not expose the wrong option.

## **What Needs to Change**

We close the analysis with an agenda for capability research. These are not predictions; they are hypotheses about where progress would be most useful.

### **A durable latent canvas.**

The single capability most likely to dissolve the three failure modes is a model-internal representation that tracks object identity, part ownership, contact, occlusion, and mark

history across generation and revision. Today’s systems approximate this with files and helper APIs. A native representation would let repair operate on parts, not pixels.

### **Tool-mediated, eventually hybrid.**

We do not believe drawing should be a fully native modality in the short term. Tools expose actions and make failures inspectable. The right long-term shape is hybrid: text/code planning, latent scene graph, differentiable rendering feedback, and image-space refinement, with tool invocations as the audit trail.

### **Training data of process, not products.**

Pretraining likely contains SVG snippets, canvas examples, ASCII diagrams, and image captions, but not enough aligned *process traces* showing how human marks become coherent bodies, props, captions, and repairs over time. The data that would unlock progress is paired storyboard-to-stroke datasets, repair traces before and after critique, and frame-by-frame animation construction data.

### **Decomposed evaluation.**

Drawing should be evaluated as a *decomposed* capability: spatial reasoning, motor planning, symbolic representation, visual memory, text layout, style control, and repair. A single Likert score on “how good is the drawing” collapses signal that researchers and tool designers both need.

### **A risk worth naming.**

Improving stroke-channel drawing quality has a provenance cost. If models produce convincing *process traces*, not only final pixels, distinguishing synthetic from human drawing becomes harder. Watermarking and provenance work that targets final images may not generalise to recorded stroke sequences.

## **A Benchmark Agenda**

### ***Why a new benchmark***

Existing text-to-image benchmarks grade final image alignment, often via CLIP similarity or human preference. These are appropriate for their original purpose but under-discriminating for the failures we describe. Two models can score similarly on final-image alignment while differing sharply in how they handle compositional constraints and repair.

We propose grading on a decomposed rubric (Table 1) and report results per dimension and per failure tag rather than as a single number.

Proposed decomposed rubric. Each dimension is scored  $\{0, 1, 2\}$ . Binary failure tags (e.g. *patch reflex observed*, *guide mark promoted*) are recorded alongside.

| Dimension        | What it measures                                                           |
|------------------|----------------------------------------------------------------------------|
| Recognisability  | Whether the intended subject is identifiable.                              |
| Spatial relation | Whether stated relations hold (left of, on, holding).                      |
| Part attachment  | Whether parts connect to their owners (hand to wrist, sleeve to shoulder). |
| Text readability | Whether captions and bubbles are legible and correctly owned.              |
| Repair quality   | Whether a critique produces structural fix or patch.                       |
| Continuity       | Whether objects persist across frames without drift.                       |

### Three benchmark tasks

We propose three tasks chosen because they discriminate on the dimensions above. Each is framed as a user-facing storyboard prompt that any INKY-like scaffold could accept; nothing in the tasks is tied to a specific preloaded example.

#### T1: Person holding a mug.

Prompt: “Animate a seated person holding a mug at a table.” Grade wrist attachment, finger-handle contact, mug-on-table contact, and posture coherence. Stresses part attachment and contact reasoning.

#### T2: Two speakers, crossed bubbles.

Prompt: “Animate two characters having a conversation in which the natural bubble layout would require tails to cross.” Grade text wrapping and tail ownership. Stresses

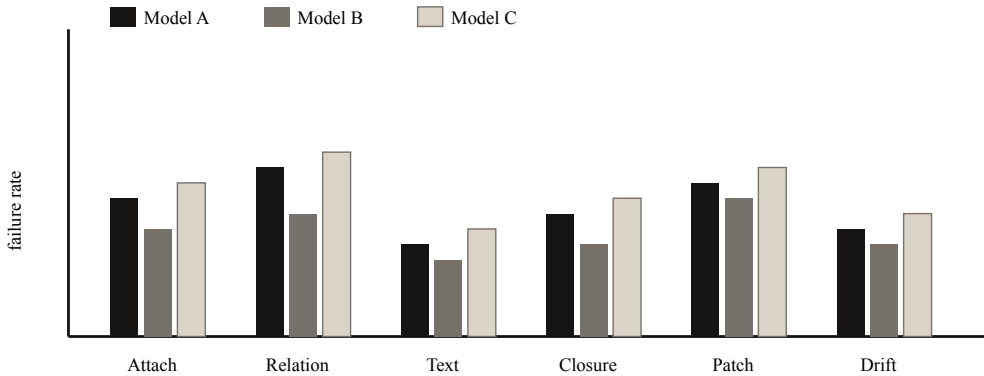
spatial relation and text readability under constraint.

#### T3: Five-frame ball handoff.

Prompt: “Animate a ball handoff across five frames.” Grade object persistence, contact at the moment of handoff, and absence of residual guide marks. Stresses continuity and *patch reflex* detection (guide marks are the canonical patch failure).

#### A planned empirical study

A full study is deferred to a follow-up. We will release the prompt suite, generated code, rendered outputs, and scoring sheets as supplementary material. Candidate models include current Claude, GPT, and Gemini variants and at least one open-weight coding-capable model; exact versions will be frozen at evaluation time. Figure 2 shows the form of the headline result we expect to report.



Illustrative form of the planned headline figure: failure rates by category across models. Values shown are placeholders; the empirical study is deferred to a follow-up.

### Related Work

Our framing draws on three strands. *Text-to-image evaluation* grades alignment of generated images to prompts and is the closest neighbour in spirit, although it operates on pixel synthesis rather than process. *Vector and symbolic visual representation* evaluates LLMs on SVG and related symbolic outputs, and is directly relevant to our claim that emitting valid markup is not the same as drawing. *Sketch generation and sketch datasets* study sketching as a generative act, often using strokes as the modelling unit; we adopt the stroke-as-process view but pair it with agentic repair, which most prior sketch work does not address.

We are gently positioning against a casual claim, common in practitioner discourse, that LLMs “can draw” because they can emit SVG or Canvas snippets that render to recognisable images. We believe this confuses fluent emission with reliable construction. We are building on, not opposing, the body of work that takes drawing seriously as a structured, evaluable behaviour.

### Limitations

This paper reports observations from a single production pipeline. The phenomena we describe are recurrent in that pipeline, but the empirical study that would let us claim rates of occurrence across models is deferred. The taxonomy



is a working taxonomy and is likely to evolve; we expect at least one additional named failure to emerge once the cross-model study is run. Finally, our scope excludes photorealistic synthesis and 3D; readers working in those areas will find some claims do not transfer cleanly.

### Conclusion

A model that can describe a hand is not yet a model that can keep the wrist attached. We have argued that current LLMs lack a durable latent canvas: a representation of object identity, part ownership, contact, and mark history that survives generation and revision. INKY’s working pipeline compensates with external structure — construction artifacts, visual feedback, and a strict *root-cause redraw* rule — and this compensation works well enough to ship animations. But the right long-term fix is internal, not external. Until models can repair drawings the way they repair code, we should evaluate them as we have proposed: on process, structure, and repair behaviour, not only on the final image.

### Acknowledgements

The author thanks the practitioners and reviewers whose feedback on the INKY pipeline shaped the taxonomy presented here, and the broader research community working on multimodal evaluation, vector representations, and sketch generation, whose work this paper builds on.

99 Saharia, C. et al. *Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding* (Imagen / DrawBench). NeurIPS, 2022.

Ghosh, D. et al. *GenEval: An Object-Focused Framework for Evaluating Text-to-Image Alignment*. NeurIPS, 2023.

*SVGenius: Benchmarking LLMs on Vector Graphics Understanding and Generation*. 2024.

*VCode: Code-as-Visual Representation for Language Models*. 2024.

Frans, K., Soros, L. B., Witkowski, O. *CLIPDraw: Exploring Text-to-Drawing Synthesis through Language-Image Encoders*. NeurIPS, 2022.

Ha, D., Eck, D. *A Neural Representation of Sketch Drawings*. ICLR, 2018.

Sangkloy, P. et al. *The Sketchy Database: Learning to Retrieve Badly Drawn Bunnies*. SIGGRAPH, 2016.

### Failure mode quick reference

| Phenomenon             | Signature                                                                         | Common trigger                                                         |
|------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Stroke amnesia         | Construction marks become final objects; speech-bubble tail tracks wrong speaker. | Lack of durable mark ownership across the generation loop.             |
| (inter-frame variant)  | Correct keyframe attachment; broken attachment in the intervening frames.         | Sequence-level loss of identity even when single frames are correct.   |
| Compositional collapse | Relational failures rise non-linearly with constraint count.                      | Multiple simultaneous spatial relations and contacts.                  |
| Patch reflex           | Defects covered (white plugs, opacity tricks, texture) rather than rebuilt.       | Critique phrased as “fix this” without anchoring back to construction. |

### Prompt suite (preview)

The following prompts will be used in the planned empirical study. Final wording and rendering targets will be frozen at evaluation time and released with the dataset.

1. Cat to the left of a kettle on a table.
2. Person wearing shorts, seated on a chair, both legs visible.
3. Two characters in conversation, both bubbles visible and correctly owned.
4. Hand holding a mug by its handle, fingers visible through the handle.
5. Office scene with five labelled sticky notes.
6. Ball moving through three frames without rendering guide arrows.
7. Thought bubble and speech bubble in the same panel, distinguishable.
8. Plant in pot: rim, soil, stem, leaves, and a water spill.

### Reproducibility note

The INKY pipeline described in this paper is implemented in JavaScript over Canvas 2D, with a node-based tool layer for rendering and review (Node  $\geq 22$ , Vite for the browser runtime). Reproducibility is at the project level rather than the example level: from any user-provided storyboard, the pipeline generates a fresh project scaffold containing requirements, source lighthouses, construction blueprints, in-between plans, semantic review boards, visual diffs, and frame schedules, all in per-project folders. The full set of pipeline stages is exposed as `npm run storyboard:*` commands (`start`, `blueprint`, `inbetween`, `captions`, `speech-bubbles`, `polish`, `visual-diff`, `semantic`, `review`, `materials`, `render-frames`). No specific example projects are preloaded as part of the reproducibility surface. The companion data release accompanying the planned empirical study will include the prompt suite, generated code per model, rendered outputs, and scoring sheets, each produced through this generic pipeline rather than against a fixed showcase. The repository, including this paper, is available at <https://github.com/poonamsnair/inky>.

1. Correspondence: [poonnair@gmail.com](mailto:poonnair@gmail.com). Melbourne, Australia. ↩